

Homework 2

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I tested and trained four different machine learning models using sklearn's linear regression model in python. Each model was responsible for trying to predict the MPG of a car from four different data points: Displacement, cylinders, weight, and acceleration. The models were trained on the automobile data provided to us in Auto.csv. The data was split into two categories, 60% of it went to training data, and the other 40% was testing data. The results of the models varied.

Feature	Slope	Intercept	MSE	R ²	t-statistic
Displacement	-0.06088	35.59	18.98	0.6512	-20.44
Acceleration	-0.1934	26.82	55.27	-0.01537	-1.009
Weight	-0.0009576	26.82	56.71	-0.04174	-1.494
Cylinders	-0.2714	25.31	56.03	-0.0293	-0.8846

Table 1: Linear Regression Model Parameters and Performance Results

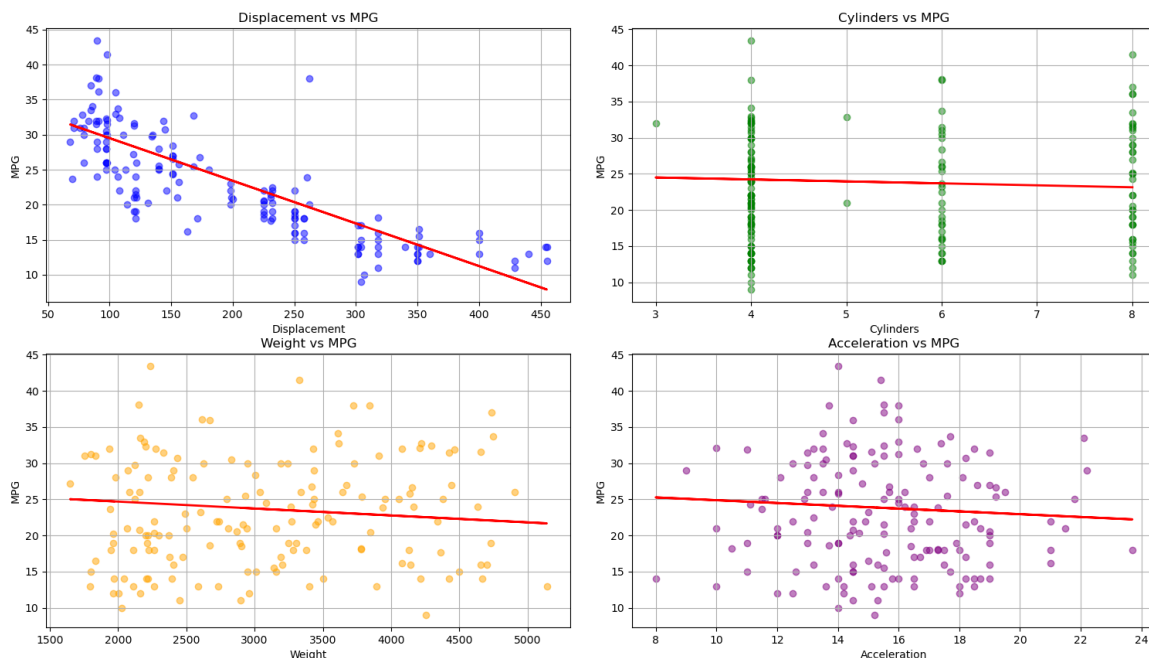


Figure 1: Linear Regression Results

The whole process went as I expected. Loading the data and training the models was straightforward and went smoothly. I referenced our in class example to help me with getting the data in the right format for the models.

Conclusion: Based on the performance statistics of the models, displacement is clearly the most

effective feature for predicting a car's MPG among the four variables tested. It performed much better in all three test statistics, MSE, R^2 and a t-statistic. Displacement's t-statistic was -20.44, which is very strong evidence against H_0 : there is no correlation between displacement and MPG. Displacement's MSE and R^2 value back this up, and I can conclude that there is a relationship between displacement and MPG.

As for the other three features, none of them had a t-statistic with a magnitude over 2, which means there isn't enough evidence to reject the null hypothesis for them. Their MSE and R^2 values were poor compared to displacement, which confirms that there isn't really a strong linear relationship between those variables and MPG.